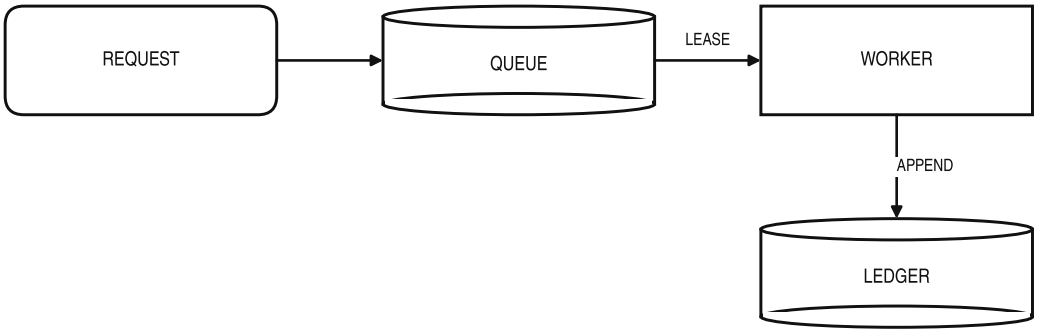
CORE BLOCKS

A worked example of the five blocks that organize a long document. Renders to paper with sheet.py and to the screen with web.py.

1 FUNCTIONAL DESCRIPTION.

A long document reads best in parts. Each part opens with a thesis and lists its sections, every section carries a status, the system map links each box to the part that explains it, a trace walks one request through the whole system, and the glossary defines each term once. The first use of a term, such as a lease or the ledger, gets a card on the screen.



CORE-501A

Figure 1-1. System Map

PART I

INTAKE

A request is accepted once, written down before anything acts on it, and never lost after that.

I.1	The front door	BUILT
I.2	The queue	PROTOTYPED

I.1 THE FRONT DOOR. BUILT

Requests arrive over one socket. The door checks the shape, assigns an id and hands the request to the queue. It answers the caller at once; the work happens later.

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Field	Meaning
id	assigned at the door, never reused
body	the request as sent

I.2 THE QUEUE. PROTOTYPED

A worker takes a request by writing a lease with an expiry; a lease that runs out returns the request to the queue. See the trace for the whole path.

Table 1-1. One Request, Door to Ledger

Step	What happens	See
1	**accept.** the door checks the shape and assigns an id	I.1
2	**enqueue.** the request is written to the queue before the caller hears back	I.2
3	**lease.** a worker writes a lease with a five minute expiry	I.2
4	**work.** the worker runs the request; a crash lets the lease expire	II.1
5	**append.** the result is appended to the ledger, and the lease is released	II.2

PART II

WORK

Workers judge; the ledger keeps the record. Nothing that moves work forward lives only in a worker's head.

II.1	The worker	DESIGNED
II.2	The ledger	BUILT
II.3	Glossary	

II.1 THE WORKER. DESIGNED

CORE

II.1.1 Retries. SPECULATIVE

A failed request goes back once with a note; a second failure goes to a person.

II.2 THE LEDGER. BUILT

Every result is one appended line, in order. Status is derived from the ledger, never kept by hand.

NOTE

A section without a status is fine where it describes no mechanism, like this example's glossary.

II.3 GLOSSARY.

Term	Definition
lease	a claim on one request with an expiry; when it runs out, the request returns to the queue
ledger	the append-only record of every result, in order
thesis	the one or two sentences that open a part and say what it argues